Day 4: Network Cable Types and ToolsTheory (30 min):Types of cables (Ethernet, fiber optics) and tools (RJ45 crimper).Practical (30 min):Create a straight-through Ethernet cable

Tasks – 1

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

➔Ethernet cable (copper)

- Signal: Electrical signals through copper wires.
- Distance: Common twisted-pair Ethernet links support up to 100 metres under standard channel specifications.
- Cost: Usually affordable and easy to install.
- Example: Connecting a desktop PC to a switch in a home or gaming café.

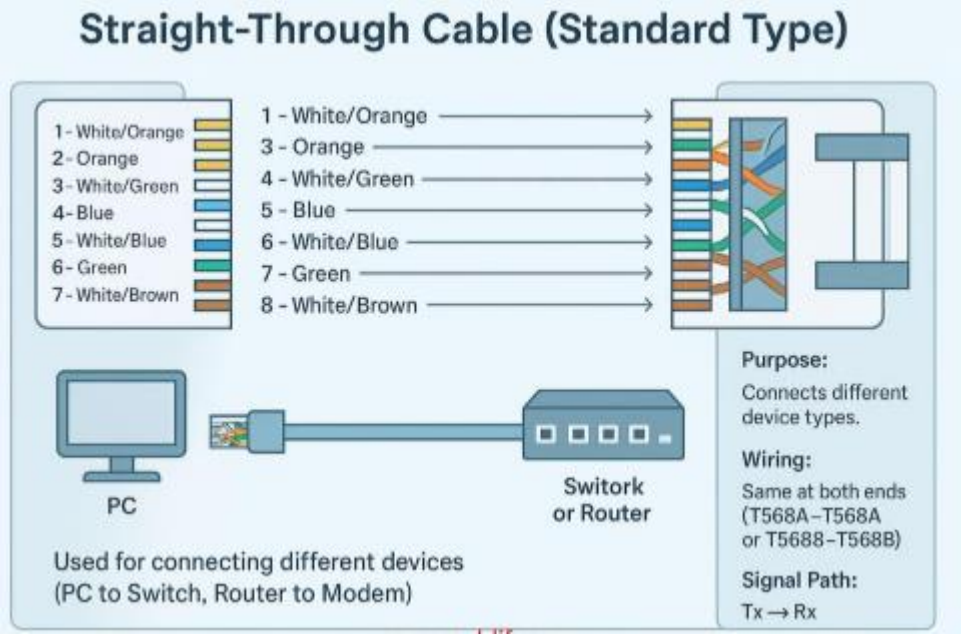
Fiber-optic cable

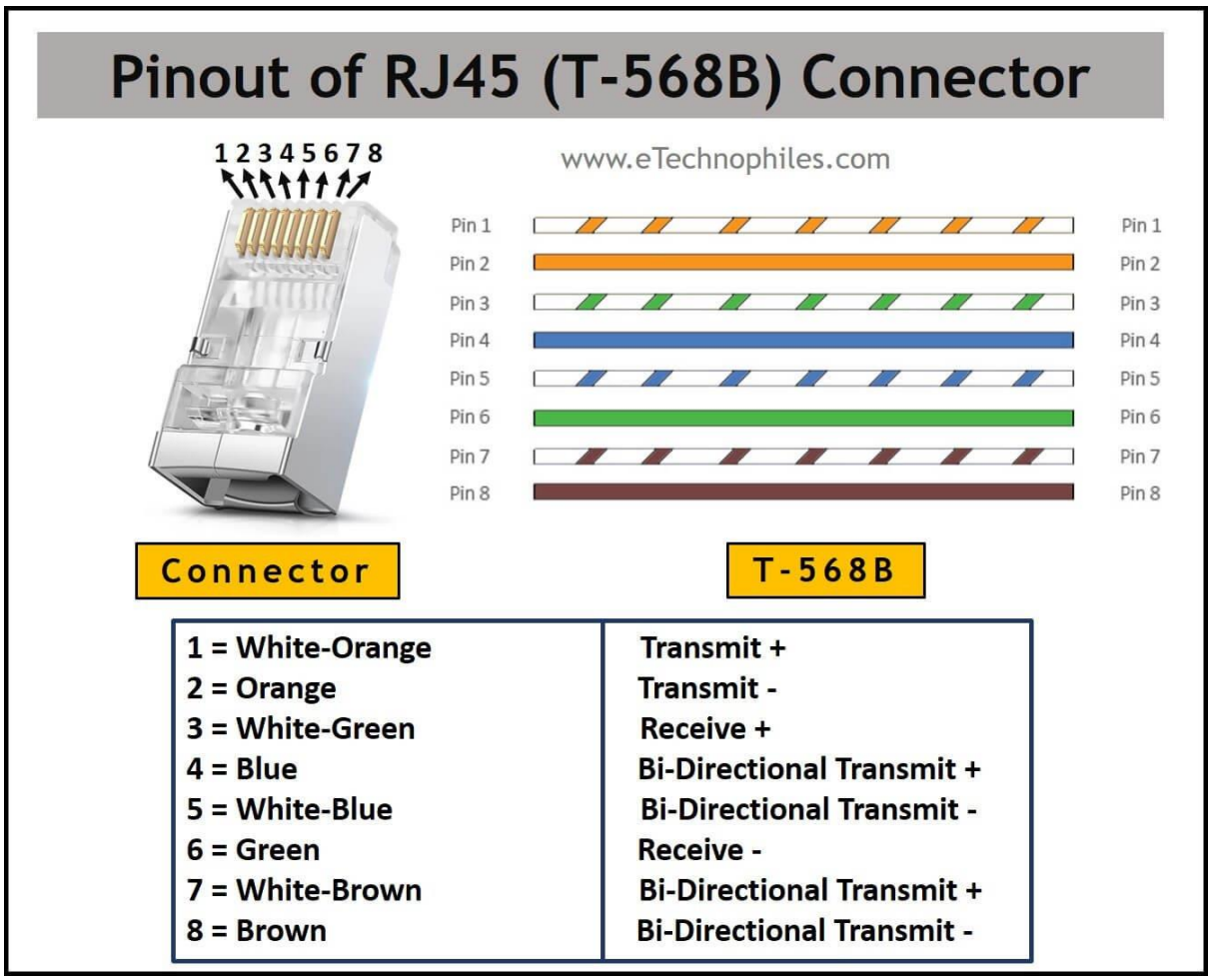
- Signal: Light travels through glass or plastic optical fibers.
- Distance: Can carry data over much longer distances, depending on the fiber type and equipment.
- Cost: Fiber equipment and installation can cost more, especially when specialist tools are required.
- Example: Connecting buildings across a college campus or linking data-center network equipment.

TASKS - 2

Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

➔

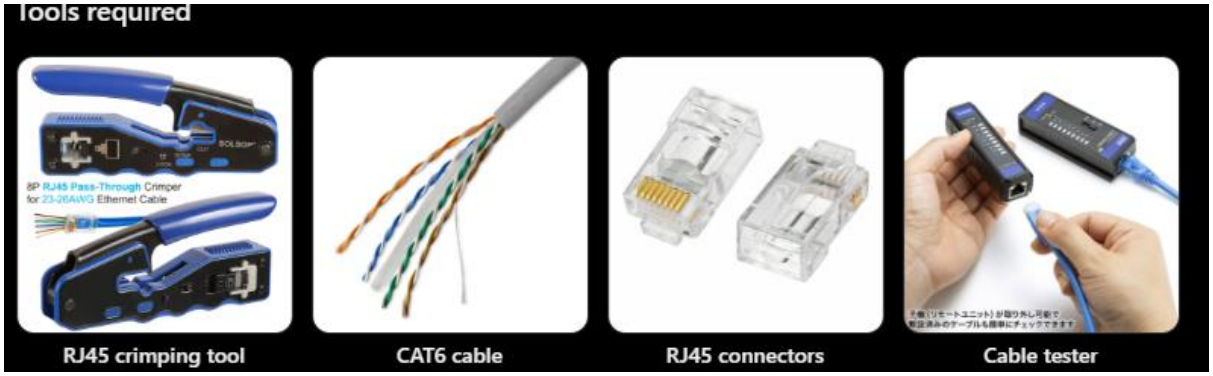


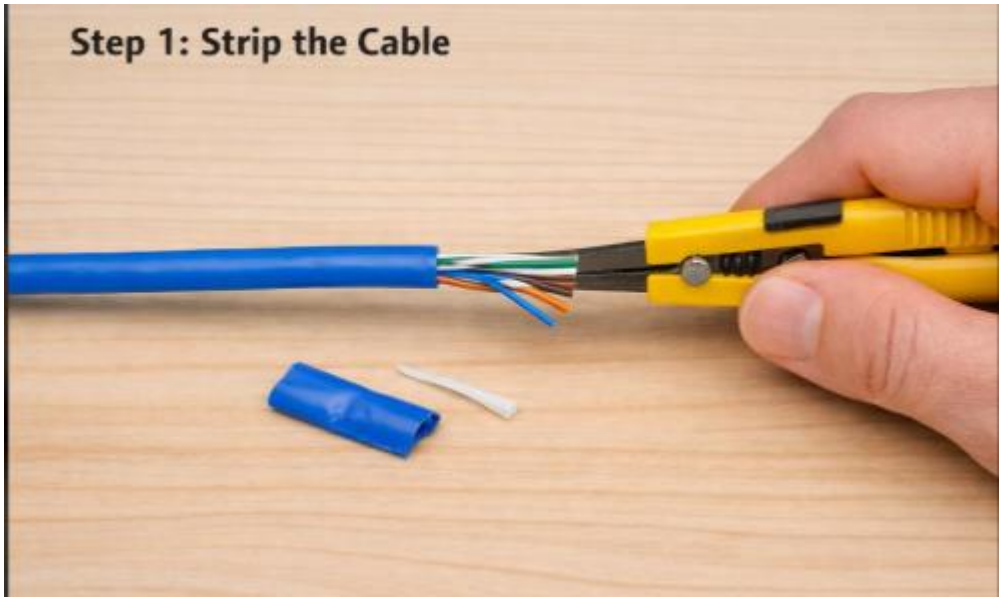


TASKS -3

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

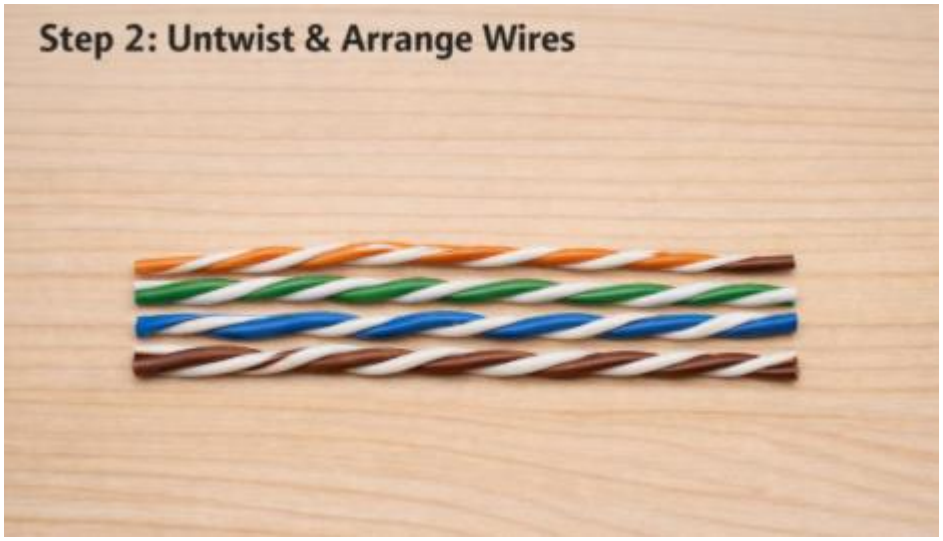
➔





Step 1: Cut and strip the cable

Cut the CAT6 cable to the required length. Carefully remove about 2–3 cm of the outer jacket without damaging the insulation of the inner wires.



Step 2: Separate the wires

Gently untwist the four pairs and straighten the eight wires. Keep the untwisted sections as short as possible to maintain cable performance.

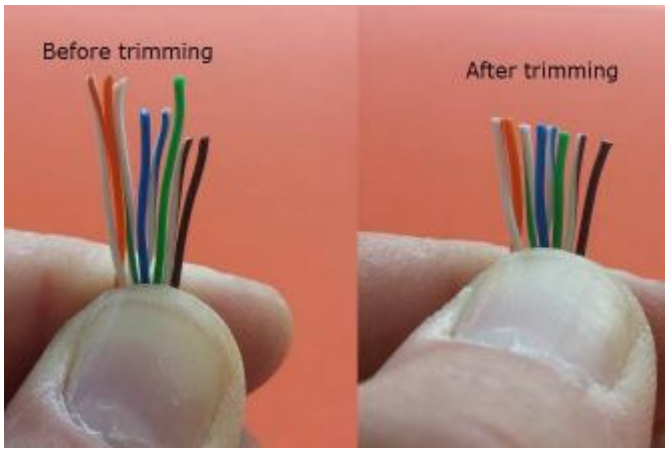
RJ45 Pin #	Wire Color (T568B)	Wire Diagram (T568B)
1	White/Orange	
2	Orange	
3	White/Green	
4	Blue	
5	White/Blue	
6	Green	
7	White/Brown	
8	Brown	

Step 3: Arrange the wires in T568B order

Arrange the wires in this exact sequence:

1. White/Orange
2. Orange
3. White/Green

4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown



Step 4: Trim the wires

Hold the wires firmly in the correct order and cut their ends evenly using the cutting section of the tool.

Step 5: Insert the wires into the RJ45 connector

Insert all eight wires into the compatible connector in the correct order. Check that every wire reaches the front of the plug and that the outer cable jacket extends inside the connector.

Step 6: Crimp the connector

Place the connector in the RJ45 crimping tool and squeeze the handles firmly to secure the contacts and cable.

Step 7: Test the cable

Connect both ends to an Ethernet cable tester. Check that the pins connect in the correct order and that there are no open circuits, shorts, or crossed wires.

Conclusion

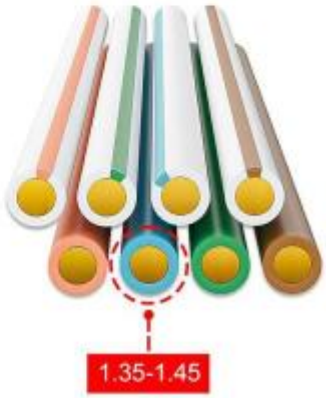
The RJ45 crimping process connects a CAT6 cable to an Ethernet connector using the T568B wiring standard. Correct wire ordering, proper crimping, and cable testing help ensure a reliable network connection between computers, routers, and switches.

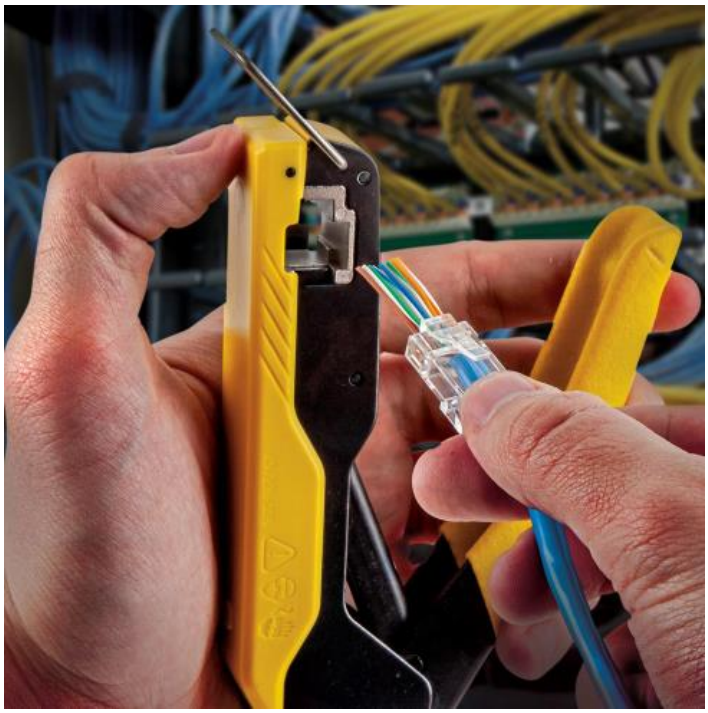




T568B

① Orange+White
② Orange
③ Green+White
④ Blue
⑤ Blue+White
⑥ Green
⑦ Brown+White
⑧ Brown





TASKS – 4

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

➔

I would choose Ethernet cables to connect the 10 PCs in my gaming café because they provide fast, stable, and low-latency connections for LAN gaming. Ethernet is more affordable than fiber optics and works well over the short distances between computers and a network switch. I would use an RJ45 crimper to attach RJ45 connectors to the Ethernet cables. This setup is budget-friendly, easy to install, and suitable for a gaming café.